

Vector Differentiation: Determining Constants in Solenoidal and Irrotational Fields

1. Mathematical Conditions Review

Name of Concepts

- Solenoidal Vector Field Condition (Divergence = 0)
- Irrotational Vector Field Condition (Curl = 0)

Core Mathematical Formulas

A vector field $\vec{F} = f_1\vec{i} + f_2\vec{j} + f_3\vec{k}$ is **solenoidal** when:

$$\nabla \cdot \vec{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} = 0$$

A vector field $\vec{F}$ is **irrotational** when:

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \vec{0}$$

Beginner-Friendly Explanation of Operations

- Solenoidal Proof:** To prove a field is solenoidal, we calculate its divergence. Notice that the constants a , b , and c do not affect the divergence calculation because they disappear during partial differentiation with respect to x , y , and z . If the final scalar values add up to exactly 0, the field is unconditionally solenoidal.
- Finding Constants Using Curl:** When a problem states that a field *is* irrotational, it gives us a powerful hint: the curl matrix expansion must yield a net zero vector ($0\vec{i} + 0\vec{j} + 0\vec{k}$). We expand the determinant, keep our unknown constants intact, and set each individual vector component ($\vec{i}$, $\vec{j}$, and $\vec{k}$) equal to zero to form simple algebraic equations that solve for a , b , and c .

2. Step-by-Step Problem Solution

Question

Prove that the vector field $\vec{F} = (x + 2y + az)\vec{i} + (bx - 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is solenoidal, and determine the values of the constants a , b , and c if $\vec{F}$ is given to be irrotational.

Step 1: Extract the components of the vector field

Identify the scalar function components f_1 , f_2 , and f_3 :

$$f_1 = x + 2y + az$$

$$f_2 = bx - 3y - z$$

$$f_3 = 4x + cy + 2z$$

Step 2: Compute partial derivatives for divergence

$$\frac{\partial f_1}{\partial x} = \frac{\partial}{\partial x}(x + 2y + az) = 1$$

$$\frac{\partial f_2}{\partial y} = \frac{\partial}{\partial y}(bx - 3y - z) = -3$$

$$\frac{\partial f_3}{\partial z} = \frac{\partial}{\partial z}(4x + cy + 2z) = 2$$

Step 3: Sum the derivatives to evaluate divergence ($\nabla \cdot \vec{F}$)

$$\nabla \cdot \vec{F} = 1 + (-3) + 2$$

$$\nabla \cdot \vec{F} = 1 - 3 + 2 = 0$$

Since the total divergence is 0, the vector field $\vec{F}$ is proven to be **solenoidal**.

Step 4: Set up the curl determinant for the irrotational condition

Because the problem specifies that the field is irrotational, we set the curl matrix equal to the zero vector:

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x + 2y + az) & (bx - 3y - z) & (4x + cy + 2z) \end{vmatrix} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

Step 5: Expand the matrix determinant component by component

For the $\vec{i}$ component:

$$\text{Component } \vec{i} = \vec{i} \left[\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z) \right]$$

$$\text{Component } \vec{i} = \vec{i}[c - (-1)] = \vec{i}(c + 1)$$

For the $\vec{j}$ component:

$$\text{Component } \vec{j} = -\vec{j} \left[\frac{\partial}{\partial x}(4x + cy + 2z) - \frac{\partial}{\partial z}(x + 2y + az) \right]$$

$$\text{Component } \vec{j} = -\vec{j}[4 - a]$$

For the $\vec{k}$ component:

$$\text{Component } \vec{k} = \vec{k} \left[\frac{\partial}{\partial x}(bx - 3y - z) - \frac{\partial}{\partial y}(x + 2y + az) \right]$$

$$\text{Component } \vec{k} = \vec{k}[b - 2]$$

Step 6: Assemble the complete expanded curl vector equation

$$\nabla \times \vec{F} = \vec{i}(c + 1) - \vec{j}(4 - a) + \vec{k}(b - 2) = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

Step 7: Match individual components to calculate constants

Equate the $\vec{i}$ scalar component to zero:

$$c + 1 = 0$$

$$c = -1$$

Equate the $\vec{j}$ scalar component to zero:

$$-(4 - a) = 0$$

$$4 - a = 0$$

$$a = 4$$

Equate the $\vec{k}$ scalar component to zero:

$$b - 2 = 0$$

$$b = 2$$

Final Answer

The field is verified as solenoidal because $\nabla \cdot \vec{F} = 0$. The required constant values making the field irrotational are:

$$a = 4$$

$$b = 2$$

$$c = -1$$